

A Consolidated Partial Answer to the Smooth Non-Complete Norm Question

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Status

This packet records the consolidated likely valid partial results for the question in Azagra–Dobrowolski–Garcia-Bravo [1]. The result is not a full solution of the question as stated. It includes the earlier separable compact-operator argument and the stronger theorem proving the desired conclusion for every infinite-dimensional Banach space admitting a bounded linear injection into some $c_0(\Gamma)$, hence for every infinite-dimensional Banach space with a Markušević basis.

Source Question

The source paper asks, on page 13 immediately before Lemma 2.10:

It is not known whether every infinite-dimensional Banach space with a C^1 equivalent norm possesses a C^1 smooth non-complete norm.

The same passage notes that for C^k , $k \geq 2$, the corresponding answer is positive by work of D'Alessandro and Hajek [2].

SMOOTH EXTRACTIONS AND APPROXIMATIONS WITHOUT CRITICAL POINTS

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It is not known whether every infinite-dimensional Banach space with a C^1 equivalent norm possesses a C^1 smooth non-complete norm.¹ The following lemma shows that for every Banach space with a C^p smooth norm there exists a kind of C^p *asymmetric non-complete subadditive functional* which successfully replaces the smooth non-complete norm in Bessaga's technique [14] for extracting points.

Lemma 2.10. *Let $(E_2, \|\cdot\|)$ be an infinite-dimensional Banach space which admits a (not necessarily equivalent) C^p smooth norm, where $p \in \mathbb{N} \cup \{\infty\}$. Then there exists a continuous function $\omega : E_2 \rightarrow [0, \infty)$ which is C^p smooth on $E_2 \setminus \{0\}$ and satisfies the following properties:*

- (1) $\omega(x + y) \leq \omega(x) + \omega(y)$, and, consequently, $\omega(x) - \omega(y) \leq \omega(x - y)$, for every $x, y \in E_2$;
- (2) $\omega(rx) = r\omega(x)$ for every $x \in E_2$, and $r \geq 0$;
- (3) $\omega(x) = 0$ if and only if $x = 0$;

Definitions

All Banach spaces are real. A norm p on a Banach space X is *continuous* if it is continuous for the original Banach norm. It is *non-complete* if the normed space (X, p) is not complete.

A bounded linear operator $T : X \rightarrow Y$ is *bounded below* if there is $c > 0$ such that $\|Tx\| \geq c\|x\|$ for every $x \in X$.

Idea of the Proof

In the separable case, countably many norm-one functionals separate points. After rapidly weighting them, they define a compact injective operator into ℓ_2 . Pulling back the Hilbert norm gives a C^∞ -smooth norm, and compactness prevents completeness.

For nonseparable spaces with coordinate structure, $c_0(\Gamma)$ is the right replacement. An infinite-dimensional subspace of $c_0(\Gamma)$ always has vectors that can be made almost supported on successively new finite coordinate sets. A positive diagonal multiplier can damp those finite sets by factors tending to zero. This multiplier remains injective on all of $c_0(\Gamma)$, but it is not bounded below on the chosen infinite-dimensional subspace.

After composing the original injection $J : X \rightarrow c_0(\Gamma)$ with this diagonal multiplier, the pullback of a C^∞ -smooth equivalent norm on $c_0(\Gamma)$ is a C^∞ -smooth norm on X . If that pullback norm were complete, the open mapping theorem would force it to be equivalent to the original Banach norm, making the composed operator bounded below. This contradicts the diagonal construction.

Theorem 1 (Separable subcase). *Let X be an infinite-dimensional separable Banach space. Then X admits a continuous C^∞ -smooth non-complete norm.*

Proof. Choose a sequence (u_n) dense in the unit sphere of X . For each n , pick $x_n^* \in S_{X^*}$ with $x_n^*(u_n) = 1$. The family (x_n^*) separates points: if $x \neq 0$, choose n with $\|u_n - x/\|x\|\| < 1/2$, and then

$$x_n^*(x/\|x\|) \geq x_n^*(u_n) - \|u_n - x/\|x\|\| > 1/2.$$

Define

$$T : X \longrightarrow \ell_2, \quad Tx = (2^{-n}x_n^*(x))_{n=1}^\infty.$$

Then T is bounded, injective, and compact, since the finite-rank truncations $T_N x = (2^{-n}x_n^*(x))_{n=1}^N$ converge to T in operator norm. Set $p(x) = \|Tx\|_{\ell_2}$. This is a continuous norm on X , and it is C^∞ -smooth on $X \setminus \{0\}$ because T is injective and the Hilbert norm is C^∞ -smooth off the origin.

If (X, p) were complete, then T would be an isometry from (X, p) onto a closed subspace of ℓ_2 . Thus $T : X \rightarrow T(X)$, viewed as a bounded bijection from the original Banach space X onto the Banach space $T(X)$, would have bounded inverse by the open mapping theorem. Hence T would be bounded below in the original norm. This is impossible for a compact operator on an infinite-dimensional Banach space. Therefore p is non-complete. $\square$

Lemma 1 (Diagonal damping in $c_0(\Gamma)$). *Let Y be an infinite-dimensional linear subspace of $c_0(\Gamma)$. Then there is a bounded injective diagonal operator*

$$D : c_0(\Gamma) \longrightarrow c_0(\Gamma), \quad (Da)(\gamma) = w_\gamma a(\gamma),$$

with $0 < w_\gamma \leq 1$, such that $D|_Y$ is not bounded below.

Proof. We inductively construct $y_n \in Y$ and pairwise disjoint finite sets $F_n \subseteq \Gamma$. Put $H_0 = \emptyset$. Having chosen $H_{n-1} = F_1 \cup \dots \cup F_{n-1}$, the space

$$Y \cap \bigcap_{\gamma \in H_{n-1}} \ker \delta_\gamma$$

is still infinite-dimensional, because only finitely many coordinate functionals have been imposed. Choose y_n in this space with $\|y_n\|_\infty = 1$. Since $y_n \in c_0(\Gamma)$, choose a finite set $E_n \subseteq \Gamma$ such that

$$|y_n(\gamma)| \leq 2^{-n} \quad (\gamma \notin E_n).$$

Set $F_n = E_n \setminus H_{n-1}$ and $H_n = H_{n-1} \cup F_n$. Then the sets F_n are pairwise disjoint, y_n vanishes on H_{n-1} , and

$$|y_n(\gamma)| \leq 2^{-n} \quad (\gamma \notin F_n).$$

Define $w_\gamma = 2^{-n}$ for $\gamma \in F_n$, and $w_\gamma = 1$ for $\gamma \notin \bigcup_n F_n$. The associated diagonal operator D is bounded and injective, since all weights are positive. For every n , the coordinates in F_n are damped by 2^{-n} , while all coordinates outside F_n already have modulus at most 2^{-n} . Hence

$$\|Dy_n\|_\infty \leq 2^{-n} \quad (n \in \mathbb{N}).$$

Thus $D|_Y$ is not bounded below. □

Theorem 2. *Let X be an infinite-dimensional Banach space. Suppose that there is a bounded injective linear operator $J : X \rightarrow c_0(\Gamma)$ for some set Γ . Then X admits a continuous C^∞ -smooth non-complete norm.*

Proof. Let $Y = JX$. By Lemma 1, choose a bounded injective diagonal operator $D : c_0(\Gamma) \rightarrow c_0(\Gamma)$ such that $D|_Y$ is not bounded below. Then $DJ : X \rightarrow c_0(\Gamma)$ is injective. It is not bounded below: if $\|DJx\|_\infty \geq c\|x\|$ for all $x \in X$, then for $y = Jx \in Y$ we would have

$$\|Dy\|_\infty \geq c\|x\| \geq \frac{c}{\|J\|} \|y\|_\infty,$$

contradicting the choice of D .

Let ρ be a C^∞ -smooth equivalent norm on $c_0(\Gamma)$. Such norms exist for arbitrary Γ , by the $c_0(\Gamma)$ smooth renorming theorem of Bible–Smith; see the discussion in Hajek–Talponen [6]. Define

$$p(x) = \rho(DJx) \quad (x \in X).$$

Since DJ is injective, p is a norm. It is continuous for the original norm because D , J , and ρ are continuous. It is C^∞ -smooth on $X \setminus \{0\}$, because $DJx \neq 0$ whenever $x \neq 0$ and ρ is C^∞ -smooth off the origin.

It remains to prove that p is not complete. Suppose, to the contrary, that (X, p) is complete. The identity map

$$I : (X, \|\cdot\|) \longrightarrow (X, p)$$

is a bounded bijection between Banach spaces, so the open mapping theorem gives a constant $C > 0$ such that $\|x\| \leq Cp(x)$ for all $x \in X$. Since ρ is equivalent to the usual $c_0(\Gamma)$ norm, there is $A > 0$ with $\rho(u) \leq A\|u\|_\infty$ for all $u \in c_0(\Gamma)$. Therefore

$$\|x\| \leq CA\|DJx\|_\infty \quad (x \in X),$$

which says that DJ is bounded below. This contradicts the construction of D . Hence p is non-complete. □

Corollary 1 (M-bases, including separable spaces). *Every infinite-dimensional Banach space with a Markuševič basis admits a continuous C^∞ -smooth non-complete norm.*

Proof. It is standard that every Banach space with a Markuševič basis admits a continuous linear injection into $c_0(\Gamma)$; see, for example, the survey discussion in Hajek–Russo [5] and the cited reference [3, Theorem II.2.4]. Theorem 2 applies. □

Consequences for the Source Problem

The source question receives a positive answer for every C^1 -smoothly renormable infinite-dimensional Banach space that also admits a bounded injection into $c_0(\Gamma)$, and in particular for every such space with an M-basis. This includes all separable spaces by the classical Markuševič theorem, and it also includes many standard nonseparable classes with M-bases or projectional skeletons. The direct compact-operator proof of Theorem 1 is kept in this consolidated packet because it is the simplest way to see the separable case.

This does not settle the full C^1 problem. The proof uses an actual coordinate injection into $c_0(\Gamma)$. A C^1 -smooth equivalent norm implies strong Asplund-type structure, but the literature search did not produce a theorem giving an M-basis, a $c_0(\Gamma)$ -injection, or a quotient smoothness principle sufficient to run this argument in every such space.

Literature and Novelty Check

Searches were run for exact phrases and close variants: “smooth non-complete norm”, “ C^1 smooth non-complete norm”, “non-complete norm Banach smooth”, $c_0(\Gamma)$ subspaces, separable compact injective operators, and Markuševič bases. The search recovered the source paper, dense smooth subspace results of Dantas–Hajek–Russo [4], c_0 -smooth renorming background [6], and M-basis surveys [5], but no exact published solution of the C^1 source question or of the c_0 -embeddable partial theorem above.

The earlier live packet `1811.07587_separable_smooth_noncomplete_norm` has been superseded by this consolidated packet. Its separable compact-operator proof is now included as Theorem 1.

Verification Notes

The argument is purely functional analytic. The key points for review are:

1. Lemma 1 only uses finite-codimensional coordinate avoidance in the infinite-dimensional subspace $Y \subseteq c_0(\Gamma)$.
2. The diagonal weights are strictly positive, so D is injective.
3. The pullback norm is smooth because DJ is injective.
4. Non-completeness follows from the open mapping theorem and the fact that DJ was constructed not to be bounded below.
5. The separable theorem is also independently verified by the compact injective ℓ_2 -operator construction.

Human Review Recommendation

Review as the single consolidated packet for this target. The theorem is not the full answer to the Azagra–Dobrowolski–Garcia-Bravo question, but it contains the separable compact-operator proof and the stronger $c_0(\Gamma)$ -embeddable theorem, hence the M-basis corollary.

References

- [1] D. Azagra, T. Dobrowolski, and M. Garcia-Bravo, *Smooth approximations without critical points of continuous mappings between Banach spaces, and diffeomorphic extractions of sets*, arXiv:1811.07587.
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- [3] R. Deville, G. Godefroy, and V. Zizler, *Smoothness and Renormings in Banach Spaces*, Pitman Monographs and Surveys in Pure and Applied Mathematics, **64**, Longman, Essex, 1993.
- [4] S. Dantas, P. Hajek, and T. Russo, *Smooth and polyhedral norms via fundamental biorthogonal systems*, Int. Math. Res. Not. IMRN **2023** (2023), 13909–13939; arXiv:2201.03379.
- [5] P. Hajek and T. Russo, *Norming Markushevich bases: recent results and open problems*, arXiv:2402.18442.
- [6] P. Hajek and J. Talponen, *Some remarks on smooth renormings of Banach spaces*, arXiv:1705.01384.